



OTC Markets Multicast Data Feeds

Technical Specification

Version 4.10.4
June 2026

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1. Introduction

1.1 Overview

This document provides a technical specification of OTC Markets Group's real-time multicast market data products. Each product provides access to a set of multicast channels carrying quote, trade and security reference data.

Each channel's data is published in duplicate on two separate multicast addresses, feed A and feed B, for high availability. Subscribers can request gap-fills or snapshots over a TCP socket-based connection.

Please visit <https://www.otcmarkets.com/market-data/real-time-data-products> for more information about these products, and for the latest version of this document.

1.2 Multicast Products

OTC Markets Group currently offers five real-time multicast Market Data products.

- **Real-Time Level 1+:** Real-time OTC Link inside quote prices (bid & offer) and real-time OTC Link Trade information. Size is aggregated for the inside price. Market Participant source attribution is not provided. Please see the [OTC Markets Display Requirements document](#) for details regarding data display.
- **Real-Time Level 2+:** Real-time quote book data including market participant information, and real-time OTC Link Trade information. The RT Level 2+ License is our premium license, which allows subscribers access to all of our multi-cast channels. Please see the [OTC Markets Display Requirements document](#) for details regarding data display.
- **Real-Time Reference Prices:** Real-time OTC Link inside quote prices (bid & offer). Quote size information is not available. The license also includes access to real-time OTC Link trade data. The innovative RTR Price license allows our market data distributors to provide this data free to their public websites and market data terminal users that want to monitor real-time OTC market prices. There is no cost to distributors of our premium products. There is a monthly fee for websites and financial portals that do not re-sell our premium market data.
- **Real-Time Level 1+ with Global OTC data:** Real-time consolidated OTC Link/Global OTC inside quote prices (bid & offer) and real-time OTC Link Trade information. Size is aggregated for the inside price. Market Participant source attribution is not provided. Please see the [OTC Markets Display Requirements document](#) for details regarding data display.
- **Real-Time Level 2+ with Global OTC data:** Real-time OTC Link/Global OTC quote book data including market participant information, and real-time OTC Link Trade information. Global OTC data is top of book data from the Global OTC ATS. Please see the [OTC Markets Display Requirements document](#) for details regarding data display.

Access to security information (tiers, caveat emptor, etc.) is provided for all five products. Access to products containing [Global OTC](#) data requires a separate data agreement with ICE.

1.3 Product-Channel Mapping

Market data is published on a set of multicast channels. A product offers access to a subset of these multicast channels. The following table shows the channels available for each product.

The **X** marks the primary channel for a given product. The **O** marks optional channels that a product subscriber may additionally choose to receive. A blank cell notes that the channel is not available for the respective product. As an example, a Level 2 + product subscriber may choose to listen only to our Quote Book and Reference Data (w cusip) channels, even though they have access to all channels.

Access to Global OTC Channels requires the appropriate licensing and agreements with both OTC Markets and ICE.

Receiving all optional channels will result in significant duplication of data.

Channel ¹ \ Product	OTC Markets Real-Time Reference Prices	OTC Markets Real-Time Level 1+	OTC Markets Real-Time Level 1+ with Global OTC data	OTC Markets Real-Time Level 2+	OTC Markets Real-Time Level 2+ with Global OTC data
Quote Book with Global OTC Channels					X
Quote Book Channels				X	O
Quote Inside with Global OTC Channels			X		O
Quote Inside Channels		X	O	O	O
Quote Reference Price Channels	X	O	O	O	O
Trade Channels	O	O	O	O	O
Reference Data Channels (w/ cusip)	O	O	O	O	O
Reference Data Channels (w/o cusip)	O	O	O	O	O

Table 1: Product-Channel Mapping

¹ For redundancy, each channel is published twice, as an A feed and a B feed.

1.4 Multicast Channels

The following table describes the available multicast channels. Each set of channels consist of two real-time data channels and two snapshot data channels.

Multicast Channel	Channel Description	Channel ID	
		Real Time	Snapshot
Quote Book Channels	All individual quote messages: price, size, qap, mmid, priceType, open/close indicator	11	12

Quote Book with Global OTC Channels	All individual quote messages: price, size, qap, mmid, priceType, open/close indicator. Includes top of book quotes from the Global OTC ATS.	19	20
Quote Inside Channels	OTC Link Best Bid and Offer: Best bid price, aggregate bid size, best offer price, aggregate offer size	14	15
Quote Inside with Global OTC Channels	Consolidated OTC Link/Global OTC Best Bid and Offer: Best bid price, aggregate bid size, best offer price, aggregate offer size	21	22
Quote Reference Price Channels	OTC Link Best Bid and Offer Prices: Best bid price, best offer price, bid and offer size set to 1. This is essentially equivalent to the Inside channel, but without the size information.	17	18
Trade Channels ¹	Trade data	1	
Extended Trade Channels ¹	Extended Trade data	49	
Reference Data Channels (with cusip) ²	Extended security (cusip, company name, etc.) information for market data recipients who own a cusip license.	5	6
Reference Data Channels (w/o cusip) ²	Extended security (company name, etc.) information for market data recipients who do not own a cusip license.	7	8

Table 2: Channel Descriptions and IDs

¹ Snapshots are not supported, Gap fill recovery should be used

² An abbreviated set of Security reference data is also sent in-line on the quote book, inside, reference price and depth channels.

2. Connectivity

This section provides a high-level overview of client integration and connection functionality.

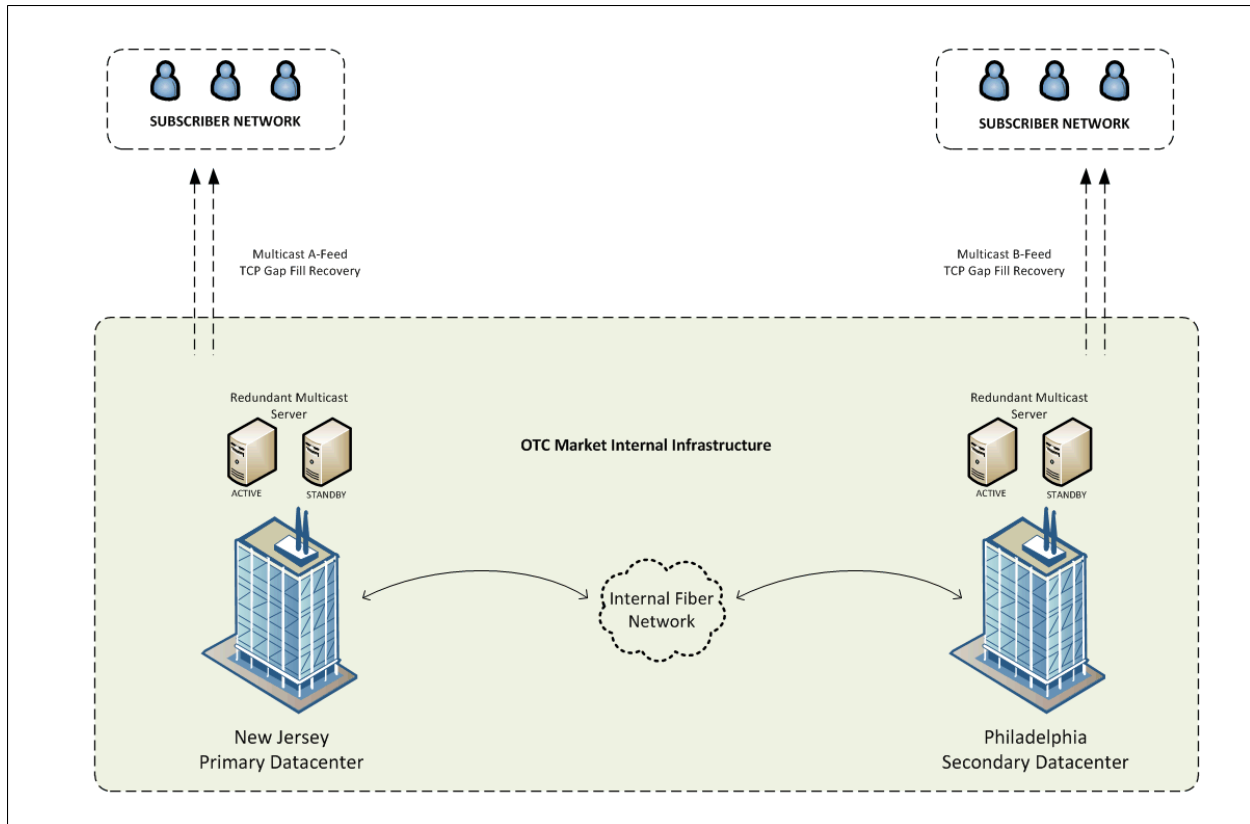
IP, Group and Port information for our production and certification environments may be found in our [Multi-cast Group documentation](#).

Clients should contact OTC Markets Group technical support or their designated network service provider to obtain technical details concerning connectivity.

2.1 Network Configuration

Because UDP is an unreliable message transport and it suffers from occasional packet loss, OTC Markets distributes each data feed using two multicast broadcasts channels. An “A” channel is published from our primary data center in Carlstadt, NJ, and a “B” channel is published from our secondary data center in Philadelphia, PA. Both data centers contain Feed Replay Servers, which can respond to Gap Fill and Snapshot requests sent by subscribers over TCP/IP. All the Feed Replay Servers contain identical information.

We have primary and standby data feed servers in our Carlstadt, NJ (A Feed) and Philadelphia, PA (B Feed) data centers. In the event of server failure in either location, the standby server will become primary and begin disseminating messages. The multicast groups and ports will remain consistent for both the primary and standby servers; however, source IPs will differ. Subscribers need to allow both the primary and standby server source IPs in order to take advantage of this redundancy. See our [multicast group document](#) for details on IPs, groups and ports.



2.2 Bandwidth Recommendations

Connection via Extranet: Please refer to <https://www.otcmarkets.com/market-data/real-time-data-products/level-1-level-2-data-products> for bandwidth requirements for each channel.

Direct Connection: Please refer to Connectivity Guide at: <https://www.otcmarkets.com/files/Connectivity-Guide-v6.1.pdf>

2.3 Supported Carriers

Please refer to for a current list of supported carriers: <https://www.otcmarkets.com/market-data/real-time-data-products/level-1-level-2-data-products>

2.4 Production and Certification Environment IP Endpoints and Multicast Groups

Please refer to <https://www.otcmarkets.com/files/binary-multi-cast-groups.pdf> for documentation on production and certification environment multicast groups, channels and source IPs.

3. Binary Channels

3.1 Binary Message Distribution

Each UDP multicast packet will contain a packet header. In the case where this header indicates it is a Heartbeat or Sequence Number Reset, the packet will contain no other messages. The SeqNum field will always contain the next expected sequence number not the current. For most packets, the PacketFlag field will be unset (zero), which indicates normal message traffic. In this case, the Messages field of the packet header will contain the number of messages contained in that packet.

Each message contained in the packet will contain a message header, which specifies the message type and message size. The message type and size fields should be used for decoding individual messages. It is important to note that future versions may append additional data fields to a message, thus proper use of the message size will be critical to ensuring backward compatibility.

Messages will be formatted in big endian, with each field having a fixed length and a fixed position.

3.1.1 Packet Header

Field	Offset	Size	Format	Description
PacketSize	0	2	Unsigned Integer	Size of packet + header size in bytes
SeqNum	2	4	Unsigned Integer	Sequence number of packet (channel specific). If heartbeat or if sequence number is being reset, will contain next expected sequence number.
PacketFlag	6	1	Bit map	see Packet flag definition
Messages	7	1	Unsigned Integer	Number of messages in packet
PacketMilli	8	4	Unsigned Integer	Milliseconds since local time midnight (EST/EDT)

3.1.2 Packet Flag

Bit	Name	Set	Clear
0	Heartbeat ¹	No message in packet	Normal message contents
1	SeqNum Reset ²	No message in packet	Normal message contents
2	Reserved		
3	Reserved		

Bit	Name	Set	Clear
4	Reserved		
5	Reserved		
6	Replay	Packet contains replay messages	Normal message contents
7	Test	Packet contains test messages. Will not occur during normal market hours	Normal message contents

¹ A Heartbeat is sent if no business level message has been published for more than a second. The heartbeats will continue to be sent in 1 second intervals until the next business level message is published.

² A SeqNumReset message will be sent at the start of day and in the scenario where a major outage leads to the feed generator application needing a fresh start. The message indicates that the channel sequence numbers are being reset to 1.

3.1.3 Message Header

Field	Offset	Size	Format	Description
MessageSize	0	2	Unsigned Integer	Size of message + header size in bytes
MessageType	2	1	Unsigned byte	See <i>Table 3: Message Type Values</i> below
Message Payload	3	-		

Message	Quote (Add, Delete, Spin)	Quote Update	Inside (Add, Delete, Spin)	Inside Update	Price Level (Add, Delete, Spin)	Price Level Update	Reference Price (Add, Delete, Spin)	Reference Price Update	Security	Start of Spin	End of Spin	Market Open	Market Close	Extended Security	Extended Security (No CUSIP)	Trade	Extended Trade
Value	1	2	3	4	5	6	7	8	9	11	12	13	14	15	16	17	18

Table 3: Message Type Values

3.2 Binary Message To Channel Mapping

Message Name	Message Type	Sent on Channel	Channel ID
Quote (Add, Delete, Spin)	1	Quote Book, Quote Book with Global OTC, Quote Book with Expert and Quote Book with Expert and Global OTC Channels	11,19,23,25
Quote Update	2	Quote Book, Quote Book with Global OTC, Quote Book with Expert and Quote Book with Expert and Global OTC Channels	11,19,23,25
Inside (Add, Delete, Spin)	3	Quote Inside, Quote Inside with Global OTC, Quote Inside with Expert, Quote Inside with Expert and Global OTC Channels	14,21,27,29
Inside Update	4	Quote Inside, Quote Inside with Global OTC, Quote Inside with Expert, Quote Inside with Expert and Global OTC Channels	14,21,27,29
Reference Price (Add, Delete, Spin)	7	Quote RT Reference Price Channel	17
Reference Price Update	8	Quote RT Reference Price Channel	17
Security Reference	9	All Quote channels	11,12, 14,15 17,18,19,21, 23,25,27,29
Start of Spin	11	All channels	
End of Spin	12	All channels	
Market Open	13	All channels	
Market Close	14	All channels	
Extended Security	15	Reference Data Channels	5
Extended Security (no CUSIP)	16	Reference Data Channels	6
Trade	17	Trade Channel	1
Extended Trade	18	Extended Trade Channel	49

3.3 Message Definitions

3.3.1 Start of Spin

This message appears on all quote channels and indicates the beginning of a spin message cycle.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level

Field	Offset	Size	Format	Description
SpinType	4	1	Unsigned byte	1 - Reference 2 - Market Data 3 - Opening
SpinStartTimeMilli	5	8	Unsigned Integer	Milliseconds since UTC epoch
SpinLastSeqNum	13	4	Unsigned Integer	Last sequence number applied to this spin
Message Size		17		

3.3.2 End of Spin

This message appears on all quote channels and indicates the end of a spin message cycle.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
SpinType	4	1	Unsigned byte	1 - Reference 2 - Market Data 3 - Opening
SpinMsgCt	5	4	Unsigned Integer	Total messages in spin
SpinEndTimeMilli	9	8	Unsigned Integer	Milliseconds since UTC epoch
SpinLastSeqNum	17	4	Unsigned Integer	Last sequence number applied to this spin
Message Size		21		

3.3.3 Market Open

This message is sent out at 6 AM on all channels. The message can also be sent mid-day if a technical outage caused a temporary closure of the market.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
MarketOpen	4	8	Unsigned Integer	Milliseconds since UTC epoch
MarketClose	12	8	Unsigned Integer	Anticipated market close Milliseconds since UTC epoch
Message Size		20		

3.3.4 Market Close

This message is sent out at 5 PM on all channels. The message can also be sent mid-day if a technical outage causes a temporary closure of the market.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
MarketCloseTimeMilli	4	8	Unsigned Integer	Milliseconds since UTC epoch midnight
MarketMsgCt	12	4	Unsigned Integer	Total day message count
Message Size		16		

3.3.5 Security

The Security Message is included in all the quote channels and provides basic security attribute information. For complete security attribute information (e.g. CUSIP), please see Extended Security Message

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
Symbol	4	10	UTF-8	Ticker symbol. Fixed income symbols may not have a symbol. For these securities, the security ID or CUSIP (Reference Data Security Message) must be used as an identifier.
LastUpdateMilli	14	8	Unsigned Integer	Milliseconds from UTC epoch
SecurityAction	22	1	Unsigned Integer	0x1 = Update 0x2 = Add 0x3= Delete 0x4= Spin
AssetClass	23	1	Unsigned Integer	0x1 = Equity 0x2 = Fixed Income
SecurityID	24	4	Unsigned Integer	Unique security ID issued by OTC Markets
SecurityFlags	28	1	BitMap	See SecurityFlag definition

Field	Offset	Size	Format	Description
Tier	29	1	Unsigned Integer	The market tier assigned by OTC Markets Group. Valid values: 0 - No Tier 2 - OTCQX U.S. 6 - OTCQX International 10 - OTCQB 20 - OTCID 21 - Pink Limited 30 - Grey Market 40 - Expert Market 50 - OTC Bonds Distributors must display with the price data, the market tier assigned in a manner acceptable to OTC Markets Group. Please see the Data Display Requirements document.
ReportingStatus	30	1	UTF-8	A - Alternative Reporting Standard B - Bank/Thrift C - SEC Reporting - Reg CF F - SEC Reporting G - International Reporting I - Insurance Company N - No Reporting O - Other Reporting Standard R - FINRA Reporting V - SEC Reporting - Investment Company W - SEC Reporting - Reg A
SecurityStatus	31	1	UTF-8	A - Active Q - Quote Only S - Suspended H - Halted I - Internal Halt R - Revoked D - Deleted
Message Size		32		

3.3.6 Security Flag

This flag notes security level attributes and is only included in the Security Message.

Bit	Name	Set	Clear
0	Proprietary Quote Eligible Flag Security is eligible for proprietary quoting under SEC Rule 15c-211.	Yes	No
1	CaveatFlag Indicates whether a Caveat Emptor warning has been applied to the security.	Yes	No
2	144A QIB Only Indicates that the security may only be purchased or sold by Qualified Institutional Buyers (QIBs) or agents acting on behalf of QIBs.	Yes	No
3	UnsolicitedOnlyFlag Indicates if security may only be quoted Unsolicited.	Yes	No
4	MultiplierFlag Indicates if multiplier applies to security quote. Multiplier data is available on www.otcmarkets.com	Yes	No
5	OTC Link ECN Eligible Indicates if security is eligible for trading on OTC Link ECN.	Yes	No
6	OTC Link Messaging Disabled	Yes	No
7	SaturationEligibleFlag Indicates if a security is eligible to have their quotes 'saturated.' Quote saturation is an OTC Link policy where quotes are removed from 'Inside Price' consideration due to a lack of responsiveness (to trade messages) by a participant.	Yes	No

3.3.7 Quote (Add, Delete, Spin)

This message is sent on the Quote Book channel and the Quote Book with Global OTC data channel

An Opening Spin is sent early in the morning, during which all quotes across all securities are disseminated. All quotes will be in Closed state at this time. Most market participants close their quotes at the end of the day, and open them again the next morning. Others delete their quotes at the end of the day, and create them anew the next morning. The quotes for the latter will not be part of the daily opening spin.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
QuoteID	4	4	Unsigned Integer	Unique Quote ID

Field	Offset	Size	Format	Description
QuoteAction	8	1	Unsigned Integer	0x2 = Add 0x3= Delete 0x4= Spin
QuoteFlags	9	1	Bit map	see QuoteFlag definition
SecurityID	10	4	Unsigned Integer	OTC Markets security ID
MPID	14	4	UTF-8	Market Participant ID owning the quote always 4 characters
AskPrice	18	8	Unsigned Integer	Price, 6 decimal places assumed
AskSize	26	4	Unsigned Integer	Number of shares
AskQAP	30	1	Integer	Specifies the access fee or rebate for the bid/offer. Positive Integers (1 to 30) indicate a rebate, and negative Integers (-1 to -30) indicate an access fee. 0 indicates no rebate or access fee.
AskTimeMilli	31	8	Unsigned Integer	Milliseconds from UTC epoch
BidPrice	39	8	Unsigned Integer	Price, 6 decimal places assumed
BidSize	47	4	Unsigned Integer	Number of shares
BidQAP	51	1	Integer	Specifies the access fee or rebate for the bid/offer. Positive Integers (1 to 30) indicate a rebate, and negative Integers (-1 to -30) indicate an access fee. 0 indicates no rebate or access fee.
BidTimeMilli	52	8	Unsigned Integer	Milliseconds from UTC epoch
QuoteReferenceID	60	2	Unsigned Integer	Numeric value (from 0 to 64,999), corresponds to FIX Tag 9670 which is assigned by the quote owner. Can be used for correlation purposes
ExtendedQuoteFlags	62	1	Bit Map	See ExtendedQuote Flag definition
Message Size		63		

3.3.8 Quote Update

Quote update information for the Quote Book channel and the Quote Book with Global OTC data channel.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
QuoteID	4	4	Unsigned Integer	Quote ID refers back to original Quote Reference
QuoteFlags	8	1	Bit map	see QuoteFlag definition
Price	9	8	Unsigned Integer	Price, 6 decimal places assumed
Size	17	4	Unsigned Integer	Number of shares
QAP	21	1	Integer	Specifies the access fee or rebate for the bid/offer. Positive Integers (1 to 30) indicate a rebate, and negative Integers (-1 to -30) indicate an access fee. 0 indicates no rebate or access fee.
QuoteTimeMilli	22	8	Unsigned Integer	Milliseconds from UTC epoch
QuoteReferenceID	30	2	Unsigned Integer	Numeric value (from 0 to 64,999) corresponds to FIX Tag 9670, which is assigned by the quote owner. Can be used for correlation purposes
ExtendedQuoteFlags	32	1	Bit Map	See ExtendedQuoteFlag definition
Message Size		33		

3.3.9 Quote Flag

This flag is part of two messages: The Quote Add/Delete/Spin message and the Quote Update message. Not all fields are applicable on every message - see table footnotes below.

Bit	Name	Set	Clear
0	Update Side ¹	Ask	Bid
1	State	Open	Closed
2	Ask Unsolicited	Unsolicited	Solicited
3	Ask Priced ²	Actual	Unpriced/BW
4	Ask BW (Bid Wanted) ²	BW	Unpriced
5	Bid Unsolicited ²	Unsolicited	Solicited

Bit	Name	Set	Clear
6	Bid Priced ²	Actual	Unpriced/OW
7	Bid OW (Offer Wanted) ²	OW	Unpriced

¹ Applicable for Update messages only. Ignore for other messages.

² Quotes can have one of three price types – Actual, Bid/Offer Wanted, or Unpriced. The Ask Price Type is represented by bits 3 and 4, and the Bid Price Type is represented by bits 6 and 7.

3.3.10 Extended Quote Flag

This flag is part of two messages: The Quote Add/Delete/Spin message and the Quote Update message.

The QuoteSaturatedFlag determines whether the quote (both bid and offer) should be considered for the inside price. If flag is set then the quote should NOT be considered for the inside.

The AutoEx Flags (Bid and Offer) note whether this quote is currently in 'AutoEx' mode. If the AutoEx flag is set for the respective quote side then a response to an OTC Link trade message will be immediate.

The National Market Securities (NMS) conditional quote flag indicates (1) the displayed quote size is a round lot at least two times greater than the minimum round lot size in the security and (2) a trade message relating to the quote cannot be sent or filled for less than the displayed size.

Bit	Name	Set	Clear
0	QuoteSaturatedFlag	Yes	No
1	BidAutoExFlag	Yes	No
2	OfferAutoExFlag	Yes	No
3	NMSConditionalQuoteFlag	Yes	No
4	Reserved		
5	Reserved		
6	Reserved		
7	Reserved		

3.3.11 Inside (Add, Delete, Spin)

This message is sent on the Quote Inside Channel and the Quote Inside with Global OTC data channel

An insideID is generated for each inside bid or offer when it is newly created. This id is unique across all inside entries for all securities, across both bids and offers. For subsequent updates and deletes, this id should be used to look up the original inside entry. For updates, the Inside Update message, defined in next section, is published.

The price of an inside bid (offer) for a security is the highest bid price (lowest offer price) of all open bids (offers) for that security. The size of an inside bid (offer) is the aggregated size of all bids (offers) at the inside bid (offer) price.

For the Quote Inside with Global OTC data channel, the inside calculation will include data from the Global OTC ATS. The inside calculation for the Quote Inside channel will only include data from the OTC Link ATS.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
InsideID	4	4	Unsigned Integer	Unique Inside ID
InsideAction	8	1	Unsigned Integer	0x2 = Add 0x3= Delete 0x4= Spin
QuoteFlags	9	1	Bit map	See Inside QuoteFlag definition
SecurityID	10	4	Unsigned Integer	OTC Markets security ID
AskPrice	14	8	Unsigned Integer	Price, 6 decimal places assumed
AskSize	22	4	Unsigned Integer	Number of shares
AskTimeMilli	26	8	Unsigned Integer	Milliseconds from UTC epoch
BidPrice	34	8	Unsigned Integer	Price, 6 decimal places assumed
BidSize	42	4	Unsigned Integer	Number of shares
BidTimeMilli	46	8	Unsigned Integer	Milliseconds since UTC epoch
AskNumPricedMP	54	1	Unsigned Integer	Number of market participants at Inside price level
BidNumPricedMP	55	1	Unsigned Integer	Number of market participants at Inside price level
Message Size		56		

3.3.12 Inside Update

Quote update information for the Quote Inside channel and the Quote Inside with Global OTC data channel. Note that Size information is aggregated across participants at the inside price level.

Saturated quotes (See Quote Message and Extended Quote Flag) will not be considered for the inside price or size. If a market participant quote which is part of the inside becomes saturated, it will be removed from the inside and new quote inside update message will be created.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
InsideID	4	4	Unsigned Integer	Inside ID refers back to original Inside Reference
QuoteFlags	8	1	Bit map	see Inside QuoteFlag definition
Price	9	8	Unsigned Integer	Price, 6 decimal places assumed
Size	17	4	Unsigned Integer	Number of shares
InsideTimeMilli	21	8	Unsigned Integer	Milliseconds from UTC epoch
NumPricedMP	29	1	Unsigned Integer	Number of market participants at Inside price level.
Message Size		30		

3.3.13 Inside Quote Flag

This flag is part of two messages: The Inside Add/Delete/Spin message, the Inside Update message. Not all fields are applicable on every message - see table footnotes below.

Bit	Name	Set	Clear
0	Update Side ¹	Ask	Bid
1	State	Open	Closed
2	Reserved		
3	Ask Priced	Actual	Unpriced
4	Ask Size OverFlow	Aggregated size exceeds 2B	Aggregated size does not exceed 2B
5	Reserved		
6	Bid Priced	Actual	Unpriced
7	Bid Size OverFlow	Aggregated size exceeds 2B	Aggregated size does not exceed 2B

¹ Applicable for Update messages only. Ignore for other messages.

3.3.14 Reference Price (Add, Delete, Spin)

This message is sent on the Reference Prices Channel. Size will always be set to 1.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
ReferencePriceID	4	4	Unsigned Integer	Unique Reference Price ID
ReferencePriceAction	8	1	Unsigned Integer	0x2 = Add 0x3= Delete 0x4= Spin
QuoteFlags	9	1	Bit map	See Reference Price QuoteFlag definition
SecurityID	10	4	Unsigned Integer	OTC Markets security ID
AskPrice	14	8	Unsigned Integer	Price, 6 decimal places assumed
AskSize	22	4	Unsigned Integer	Always 1
AskTimeMilli	26	8	Unsigned Integer	Milliseconds from UTC epoch

Field	Offset	Size	Format	Description
BidPrice	34	8	Unsigned Integer	Price, 6 decimal places assumed
BidSize	42	4	Unsigned Integer	Always 1
BidTimeMilli	46	8	Unsigned Integer	Milliseconds since UTC epoch
Message Size		54		

3.3.15 Reference Price Update

Quote update information for the Reference Price channel. Note 'Size' will always be set to 1.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
ReferencePriceID	4	4	Unsigned Integer	Unique Reference Price ID
QuoteFlags	8	1	Bit map	See Reference Price QuoteFlag definition
Price	9	8	Unsigned Integer	Price, 6 decimal places assumed
Size	17	4	Unsigned Integer	Always 1
TimeMilli	21	8	Unsigned Integer	Milliseconds from UTC epoch
Message Size		29		

3.3.16 Reference Price Quote Flag

This flag is part of two messages: The Reference Price Add/Delete/Spin message, the Reference Price Update message, Not all fields are applicable on every message - see table footnotes below.

Bit	Name	Set	Clear
0	Update Side ¹	Ask	Bid
1	State	Open	Closed
2	Reserved		
3	Ask Priced	Actual	Unpriced

Bit	Name	Set	Clear
4	Reserved		
5	Reserved		
6	Bid Priced	Actual	Unpriced
7	Reserved		

¹ Applicable for Update messages only. Ignore for other messages.

3.3.17 Extended Security

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
Symbol	4	10	UTF-8	Ticker symbol
LastUpdateMilli	14	8	Unsigned Integer	Milliseconds from UTC epoch
SecurityAction	22	1	Unsigned Integer	0x1 = Update 0x2 = Add 0x3 = Delete 0x4 = Spin
OTCIssuerID	23	4	Unsigned Integer	Unique issuer ID
SecurityDesc	27	25	UTF-8	Description
ShortName	52	25	UTF-8	Name of Security.
AssetClass	77	1	Unsigned Integer	0x1 = Equity 0x2 = Fixed Income

Field	Offset	Size	Format	Description
SecurityType	78	5	UTF-8	Indicates type of security: CS - Common Stock PS - Preferred Stock RTS - Rights UTS - Units OS - Ordinary Shares ADR - American Depository Receipts GDR - Global Depository Receipts WTS - Warrants OTHER - Other Security Type FUND - Fund NYRS - New York Registry Shs SP - Structured Product ETF - Exchange-Traded Fund CORP – Corporate Bond AGEN – Agency Bond EQLK – Equity Linked Bond
PrimaryMarket	83	3	UTF-8	Used to indicate which market or stock exchange the security primarily trades on: OP – OTC Link G – Grey Market OY – OTC Bonds N – NYSE O – NASDAQ A – NYSE American X – NYSE ARCA C – CBOE I – IEX M – Members Exchange (MEMX) L – Long Term Stock Exchange (LTSE)
SecurityID	86	4	Unsigned Integer	Unique security ID issued by OTC Markets
SecurityFlags	90	2	BitMap	See Extended SecurityFlag definition

Field	Offset	Size	Format	Description
Tier	92	1	Unsigned Integer	The market tier assigned by OTC Markets Group. Valid values: 0 - No Tier 2 - OTCQX U.S. 6 - OTCQX International 10 - OTCQB 20 - OTCID 21 - Pink Limited 30 - Grey Market 40 – Expert Market 50 – OTC Bonds Distributors must display with the price data, the market tier assigned in a manner acceptable to OTC Markets Group. Please see the Data Display Requirements document.
ReportingStatus	93	1	UTF-8	- A - Alternative Reporting Standard - B - Bank/Thrift - C – SEC Reporting – Reg CF - F – SEC Reporting - G - International Reporting - I - Insurance Company - N -No Reporting - O - Other Reporting Standard - R - FINRA Reporting - V - SEC Reporting - Investment Company - W – SEC Reporting – Reg A
Disclosure Status	94	1	Unsigned Integer	The current disclosure status of the issuer: 0 – No Disclosure Status 2 –Current Information 3 – Limited Information 4 – No Information

Field	Offset	Size	Format	Description
SecurityStatus	95	1	UTF-8	A – Active Q – Quote Only S – Suspended H – Halted I – Internal Halt R – Revoked D – Deleted
ParValue	96	8	Unsigned Integer	6 decimal places assumed. 0 should be read as null or 'not available'.
Coupon	104	8	Unsigned Integer	6 decimal places assumed. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'
MaturityDateMilli	112	8	Unsigned Integer	Milliseconds from UTC epoch. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'
CallableDateMilli	120	8	Unsigned Integer	Milliseconds from UTC epoch. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'
ADRRatio	128	8	Unsigned Integer	6 decimal places assumed. Field only applies to equity securities (see AssetClass) where security type = ADR, GDR or NYRS. 0 should be read as null or 'not available.'
ADRLevel	136	15	UTF-8	Field only applies to equity securities (see AssetClass) where security type = ADR, GDR or NYRS. 0 should be read as null or 'not available.'
Security Detail Size	151	1	Unsigned Integer	Size of Security Detail field If zero Security Detail will not be present
Security Detail	152	Variable maximum 75 (SecurityDetail Size)	UTF-8	Security Detail
Issuer Size	Varies	1	Unsigned Integer	Size of issuer name field. If zero Issuer Name will not be present
IssuerName	Varies	Variable maximum 75 (IssuerSize)	UTF-8	Issuer Name
CUSIP	Varies	9	UTF-8	
Message Size		161-324		

3.3.18 Extended Security Flag

This flag notes security level attributes and is only included in the Security Message.

Bit	Name	Set	Clear
0	Proprietary Quote Eligible Flag Security is eligible for proprietary quoting under SEC Rule 15c-211.	Yes	No
1	CaveatFlag Indicates whether a Caveat Emptor warning has been applied to the security.	Yes	No
2	144A QIB Only Indicates that the security may only be purchased or sold by Qualified Institutional Buyers (QIBs) or agents acting on behalf of QIBs.	Yes	No
3	UnsolicitedOnlyFlag Indicates if security may only be quoted Unsolicited.	Yes	No
4	MultiplierFlag Indicates if multiplier applies to security quote. Multiplier data is available on www.otcm Markets.com	Yes	No
5	OTC Link ECN Eligible Indicates if security is eligible for trading on OTC Link ECN.	Yes	No
6	OTC Link Messaging Disabled	Yes	No
7	SaturationEligibleFlag Indicates if a security is eligible to have their quotes 'saturated.' Quote saturation is an OTC Link policy where quotes are removed from 'Inside Price' consideration due to a lack of responsiveness (to trade messages) by a participant.	Yes	No
8	Investment Grade Applies to fixed income only. Indicates if the security is Investment Grade. Status source is FINRA TRACE.	Yes	No
9	TradingFlat Applies to fixed income only. Indicates if the security is trading flat (no coupon)	Yes	No
10	Callable Applies to fixed income only. Indicates if the security is callable.	Yes	No
11-15	Reserved		

3.3.19 Extended Security (No CUSIP)

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level

Field	Offset	Size	Format	Description
Symbol	4	10	UTF-8	Ticker symbol
LastUpdateMilli	14	8	Unsigned Integer	Milliseconds from UTC epoch
SecurityAction	22	1	Unsigned Integer	0x1 = Update 0x2 = Add 0x3= Delete 0x4= Spin
OTCIssuerID	23	4	Unsigned Integer	Unique issuer ID
SecurityDesc	27	25	UTF-8	Description
ShortName	52	25	UTF-8	Name of Security. If there is not ShortName, then text in the Issuer field should be used instead.
AssetClass	77	1	Unsigned Integer	0x1 = Equity 0x2 = Fixed Income
SecurityType	78	5	UTF-8	Indicates type of security: CS - Common Stock PS - Preferred Stock RTS - Rights UTS - Units OS - Ordinary Shares ADR - American Depository Receipts GDR - Global Depository Receipts WTS - Warrants OTHER - Other Security Type FUND - Fund NYRS - New York Registry Shs SP - Structured Product ETF - Exchange-Traded Fund CORP – Corporate Bond AGEN – Agency Bond EQLK – Equity Linked Bond

Field	Offset	Size	Format	Description
PrimaryMarket	83	3	UTF-8	Used to indicate which market or stock exchange the security primarily trades on: OP – OTC Link G - Grey Market OY – OTC Bonds N – NYSE O – NASDAQ A – NYSE American X – NYSE ARCA C – CBOE I – IEX M – Members Exchange (MEMX) L – Long Term Stock Exchange (LTSE)
SecurityID	86	4	Unsigned Integer	Unique security ID issued by OTC Markets
SecurityFlags	90	2	BitMap	See Extended SecurityFlag definition
Tier	92	1	Unsigned Integer	The market tier assigned by OTC Markets Group. Valid values: 0 - No Tier 2 - OTCQX U.S. 6 - OTCQX International 10 - OTCQB 20 - OTCID 21 - Pink Limited 30 - Grey Market 40 – Expert Market 50 – OTC Bonds Distributors must display with the price data, the market tier assigned in a manner acceptable to OTC Markets Group. Please see the Data Display Requirements document.

Field	Offset	Size	Format	Description
ReportingStatus	93	1	UTF-8	A - Alternative Reporting Standard B - Bank/Thrift C – SEC Reporting – Reg CF F - SEC Reporting G - International Reporting I - Insurance Company N -No Reporting O - Other Reporting Standard R - FINRA Reporting V - SEC Reporting - Investment Company W – SEC Reporting – Reg A
Disclosure Status	94	1	Unsigned Integer	The current disclosure status of the issuer: 0 – No Disclosure Status 2 –Current Information 3 – Limited Information 4 – No Information
SecurityStatus	95	1	UTF-8	A – Active Q – Quote Only S – Suspended H – Halted I – Internal Halt R – Revoked D – Deleted
ParValue	96	8	Unsigned Integer	6 decimal places assumed. 0 should be read as null or 'not available'.
Coupon	104	8	Unsigned Integer	6 decimal places assumed. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'
MaturityDateMilli	112	8	Unsigned Integer	Milliseconds from UTC epoch. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'
CallableDateMilli	120	8	Unsigned Integer	Milliseconds from UTC epoch. Field only applies to fixed income (see AssetClass). 0 should be read as null or 'not available.'

Field	Offset	Size	Format	Description
ADRRatio	128	8	Unsigned Integer	6 decimal places assumed. Field only applies to equity securities (see AssetClass) where security type = ADR, GDR or NYRS. 0 should be read as null or 'not available.'
ADRLevel	136	15	UTF-8	Field only applies to equity securities (see AssetClass) where security type = ADR, GDR or NYRS. 0 should be read as null or 'not available.'
Security Detail Size	151	1	Unsigned Integer	Size of Security Detail field If zero Security Detail will not be present
Security Detail	152	Variable maximum 75 (SecurityDetail Size)	UTF-8	Security Detail
Issuer Size	Varies	1	Unsigned Integer	Size of issuer name field. If zero Issuer Name will not be present
IssuerName	Varies	Variable maximum 75 (IssuerSize)	UTF-8	Issuer Name
Message Size		152-317		

3.3.20 Trade

The Trade Message is only sent on the Trade Channel.

A Trade Channel message is created for every trade that occurs on OTC Markets OTC Link ATS. Note: The Trade Channel does not include all OTC equity trades. It only includes those trades negotiated on the OTC Link ATS platform.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
TradeID	4	4	Unsigned Integer	Unique Trade ID
TradeAction	8	1	Unsigned Integer	0x2 = Add
TradeFlags	9	1	Bit map	see TradeFlag definition
SecurityID	10	4	Unsigned Integer	OTC Markets security ID

Field	Offset	Size	Format	Description
TradeStatus	14	1	Bit map	See TradeStatus definition
Venue	15	3	UTF-8	Executing Venue. Values include: • ATS
Deprecated	18	5	UTF-8	Deprecated. Spaces will be sent in message.
TradePrice	23	8	Unsigned Integer	Price, 6 decimal places assumed
TradeSize	31	4	Unsigned Integer	Number of shares
TradeTimeMilli	35	8	Unsigned Integer	Milliseconds from UTC epoch
Message Size		43		

3.3.21 Extended Trade

An Extended Trade Channel message differs from a Trade Channel message, in that it is created for every trade that occurs on OTC Markets OTC Link ATS, OTC Link ECN, and OTC Link NQB. Note: The Trade Channel does not include all OTC equity trades. It only includes those trades negotiated on OTC Link ATS, or executed on the OTC Link ECN or OTC Link NQB platforms. It also differs from a Trade Message, in that TradeID has 8 bytes, instead of 4.

Field	Offset	Size	Format	Description
ChannelSeqNum	0	4	Unsigned Integer	Monotonically increasing message sequence number at the channel level
TradeID	4	8	Unsigned Integer	Unique Trade ID
TradeAction	12	1	Unsigned Integer	0x2 = Add
TradeFlags	13	1	Bit map	see TradeFlag definition
SecurityID	14	4	Unsigned Integer	OTC Markets security ID
TradeStatus	18	1	Bit map	See TradeStatus definition
Venue	19	3	UTF-8	Executing Venue. Values include: • ATS • ECN • NQB
Deprecated	22	5	UTF-8	Deprecated. Spaces will be sent in message.
TradePrice	27	8	Unsigned Integer	Price, 8 decimal places assumed
TradeSize	35	8	Unsigned Integer	Number of shares, 8 decimal places assumed
TradeTimeMilli	43	8	Unsigned Integer	Milliseconds from UTC epoch
Message Size		51		

3.3.22 Trade Flag

Bit	Name	Set	Clear
0	Deprecated		
1-7	Reserved		

3.3.23 Trade Status

Bit	Name	Set	Clear
0	Irregular	Yes	No
1-7	Reserved		

4. Message Recovery

Since by its nature multicast distribution is unreliable, messages may be lost or delivered out of order. Therefore, the subscriber must implement message recovery processing. To aid in this processing three recovery mechanisms are provided:

1. Multicast Group Redundancy: The data for each product is distributed via two multicast groups (A/B) that are routed over separate network paths.
2. Gap Fills: A TCP socket based message recovery service is provided for any messages that are missed on both A and B feeds.
3. On-Demand Snapshots: Snapshots may be requested via TCP for all channels excluding the OTC Link Trade channel. Data will be delivered via the respective dedicated snapshot channel.

Note: A test/certification environment is available. Please refer to our 'Multicast Group' document at <https://www.otcmarkets.com/files/binary-multi-cast-groups.pdf> for the correct IPs/ports.

4.1 A/B Feed Arbitration

The real time data for each product is distributed via two multicast groups (A/B) that are routed over separate network paths. The A channel is published from our primary data center in Carlstadt and the B channel is published out of our secondary data center in Philadelphia.

Each of the A and B channels will contain the identical message level traffic, but not identical packet level traffic. The message level sequence number can be used to detect gaps on an individual channel. If a gap is detected on one channel, the missing messages can be recovered from the other channel. This arbitration should be done at the message level, and not at the packet level.

Note that the Snapshot channels are also published from both Data Centers as A and B channels. However, the snapshot channels are not synchronized and cannot be arbitrated. If gaps are detected on a snapshot, must wait for/request another snapshot. There is no gap fill for snapshot data.

4.2 Gap Fill Recovery

The Recovery Server listens on a TCP socket for Gap Fill requests. Subscribers can use this service for requesting resends of missed market data messages from the multicast channels. The subscriber should initiate a TCP socket connection with the Recovery Server when a message gap is detected. After the replay request has been satisfied by the Recovery Server the TCP socket will be closed by the service.

Before requesting a resend of "missed" messages, the subscriber must make sure that the particular message or messages have indeed been missed, and have not simply been delivered out of order by the underlying UDP protocol. This procedure would entail keeping track of missed messages by using a combination of techniques e.g. tracking ApplSeqNum for gaps and setting a timer at the expiry of which, if the missing message or messages has not been received it is safe to assume the message or messages are lost and no longer available on the multicast stream. Another suggestion is to set a "gap tolerance" of 'N' messages -- a resend request should only be sent after receiving the Nth message (by sequence number) after the missed message.

The Recovery Server supports replay requests for missed messages from the real-time data multicast channels. Use field 1355 (RefApplID) in the Replay Request message to specify the channel being requested.

One Recovery Server is located in our primary data center (Carlstadt) from where the A feeds are published, and one Recovery Server is located in our secondary data center (Philadelphia) from where the B feeds are published. For gaps on the real-time channels, the gap fill request may be sent to either data center.

Gap Fill Restrictions: A single request is limited to a maximum of 2000 messages. To fill larger gaps, clients will need to send multiple requests or request a snapshot (See 4.3 Snapshot Recovery). The server will enforce throttling on the resend connection to prevent excessive resend activity on one client connection from negatively affecting the overall system. Therefore, requests may be queued if received at a rapid rate. Throttles are based on market conditions and may change over time.

The Gap Fill mechanism exists to enable subscribers to recover from short network or application outages. To recover from longer outages, the Snapshot Recovery mechanism should be used.

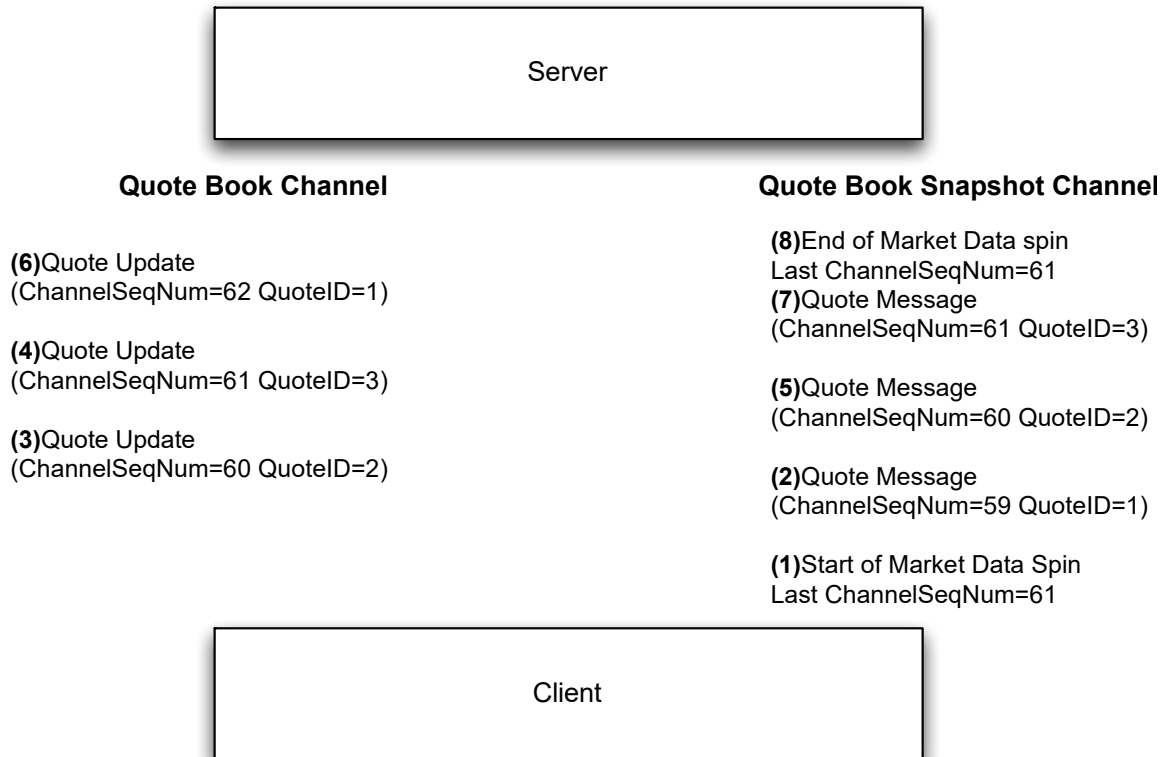
4.3 Snapshot Recovery

To assist in quick recovery after a system failure at a client, a snapshot request feature is available for the following market data channels:

1. Quote Book Channel
2. Quote Book with Global OTC Channel
3. Quote Inside Channel
4. Quote Inside with Global OTC Channel
5. Reference Price Channel
6. Extended Security (with CUSIP)
7. Extended Security (without CUSIP)

Requesting a Multicast Snapshot: Each real-time multicast feed channel has a corresponding dedicated snapshot channel. A snapshot request (Replay Request with 1347 = 1) can be sent on the TCP socket connection to the Recovery Server. On receiving the request, the Recovery Server will acknowledge the request by sending a Resend Request Ack on the TCP connection, and start publishing a snapshot on the appropriate multicast snapshot channel. The TCP connection will be closed after the Resend Request Ack has been sent. In some cases, the subscriber may start receiving the snapshot before the Ack is received. If a snapshot broadcast is in progress, the request snapshot will not begin until the in progress snapshot has completed.

Processing a Multicast Snapshot: Before sending a snapshot request message, the subscriber must buffer all messages on the appropriate product multicast channel. Once a complete snapshot refresh has been received, the subscriber/client can apply the buffered messages and then resume normal real-time message processing.



Message Processing Timeline

- (1)** Quote Book Spin begins indicating the last sequence number applied to this spin is 61
- (2)** Quote Message arrives and is processed
- (3)** Quote Update arrives and is buffered
- (4)** Quote Update arrives and is buffered
- (5)** Quote Message arrives and is processed
- (6)** Quote Update arrives and is buffered
- (7)** Quote Message arrives and is processed
- (8)** Quote Book Spin completes
- (9)** Quote Update **(3)** discarded message ChannelSeqNum < last ChannelSeqNum applied
- (10)** Quote Update **(4)** discarded message ChannelSeqNum < = last ChannelSeqNum applied
- (11)** Quote Update **(6)** applied message ChannelSeqNum > last ChannelSeqNum applied
- (12)** Recovery complete normal processing resumes

4.4 Recovery Message Definitions

The following messages are supported on the TCP socket based resend channel:

- Replay Request
- Replay Request Ack
- Appropriate market data message (for gap fill requests)

The Replay Request and Replay Request Ack messages are formatted in a TAG=VALUE[SOH] FIX-like format. i.e. each field consists of four items:

1. The tag number
2. The = sign
3. The value
4. The SOH character

The messages are terminated by the final SOH character on the checksum field.

4.4.1 Replay Request

Tag	Field Name	Required	Description
35	Message Type	Y	BW
49	SenderCompID	Y	Message Sender
1346	ApplReqID	Y	Unique ID identifying this request.
1347	ApplReqType	N	0 = Gap Fill Request 1 = Snapshot Request If this field is not present, value 0 (Gap Fill Request) is assumed
1355	RefApplID	Y	A unique id identifying the applicable channel for the request. Use IDs defined in Table 2: Channel Descriptions and IDs
1182	ApplBegSeqNo	Y	Application sequence number of first message in range to be resent. Not required for snapshot request.
1183	ApplEndSeqNo	Y	Application sequence number of last message in range to be resent. If request is for a single message ApplBeginSeqNo = ApplEndSeqNo. A maximum of 2000 messages can be requested per Gap Fill Request message. Not required for snapshot request.
10	Checksum ¹	Y	Three byte checksum

¹ Follow the standard FIX protocol algorithm in calculating the checksum. This consists of summing up the decimal value of the UTF-8 representation all the bytes up to the checksum field (which is last) and returning the value modulo 256.”

4.4.2 Resend Request Ack

Tag	Field Name	Present	Description
35	Message Type	Always	BX
59	TargetCompID	Always	Message Recipient
1346	ApplReqID	Always	Identifier of the request associated with this ACK message
1348	ApplResponseType	Always	0 – Request successfully processed 1 – Request limits exceeded 2 – Messages are not available 3 – User not entitled to application 4 – Badly formed request Field 58 may provide additional details.
58	Text	Sometimes	May contain additional descriptive detail about the response when 1348 is non-zero.
1355	RefApplID	Always	Echo back of the RefApplID received in the Request message.
1182	ApplBegSeqNo	Sometimes	Application sequence number of first message in range to be resent. Present if field 1348 = 0.
1183	ApplEndSeqNo	Sometimes	Application sequence number of last message in range to be resent. Present if 1348 = 0.
10	Checksum	Always	Three Byte Checksum

If the Replay Request was for a Snapshot, the TCP socket connection will be terminated by the Recovery Server after the Replay Request Ack is sent.

If the Replay Request was for a Gap Fill, the appropriate messages will follow the Replay Request Ack.

Please use our replay server test/certification environment for testing. Details regarding the test environment may be found in our 'Multicast Group' document at <https://www.otcmarkets.com/files/binary-multi-cast-groups.pdf>.

Document Revision History

Version	Description of Version	Date Completed
4.10.4	Changed Extended Trade Channel ID from 32 to 49.	2026.06.16
4.10.3	Changed Extended Trade Message TradePrice to 8 decimal places assumed Corrected Extended Trade message TradeSize to be 8 decimal places assumed.	2026.05.26
4.10.2	Changed Extended Trade Message TradeSize from 4 bytes to 8 bytes and indicated 6 decimal places assumed for fractional shares	2026.03.12
4.10.1	Added Expert channels to Binary Message To Channel Mapping.	2026.03.04
4.10	Added Extended Trade channel and message.	2026.03.03
4.9.1	Corrected Trade Message Venue byte size from 4 to 3.	2025.09.11
4.9	Removed separate OTC Link ECN Trade channel. Added Venue to Trade message to indicate execution venue: ATS, ECN or NQB.	2025.09.11
4.8	Tier name change from 'Pink Current' to 'OTCID.' Deprecation of OTCQX US Premier and OTCQX International Premier tiers.	2025.05.02
4.7	Added the OTC Link ECN Trade channel.	2023.07.27
4.6	Removal of BB Quoted Flag / Replaced with 'MultiplierFlag' on Security Message. Removal of OTCBB Only Tier and Pink No Information Tier.	2022.11.21
4.5	Addition of Reporting Standard value (SEC Reporting – Reg CF)	2022.07.28
4.4	Label Change: Proprietary Quote Eligible replaces 'Piggyback' Flag field label.	2021.05.17

4.3	Addition of Primary Market Venues: BATS, IEX, MEMX, LTSE	2019.12.11
4.2	Addition of Expert Market Tier (40) to Tier Field in Security Message	2019.04.18
4.1	Replaced Reg SHO flag with 144A/QIB Only Flag	2018.08.20
4.0	Removed Trade Statistic message from Trade Channel	2018.03.23
3.9	Updated ASCII fields to UTF-8	2018.01.25
3.8	Replaced 144A flag with OTCM ECN Eligible flag	2017.07.28
3.7	Amendment of UnsolicitedOnlyFlag definition.	2017.05.26
3.6	Addition of Quote Book and Quote Inside Channels containing Global OTC data	2016.12.19
3.5	Updated delay intervals for Trade Statistics message	2016.07.01
3.4	Clarified 'Saturation' concept for Inside messages	2016.06.09
3.3	Addition of Trade Statistic message, Addition of ExtendedQuoteFlag to Quote message, Addition of SaturationEligible Flag to Security Flag	2015.10.01
3.2	Updates for server redundancy	2015.04.29
3.1	Add State Open/Close flag for Insides and Reference Price	2013.12.09
3.0	New Binary Messages; removal of Price Depth Channel documentation	2013.07.11
2.4	Clarification of packet header timestamp Removed details regarding periodic snapshots	2012.10.09
2.3	Clarification regarding null symbols for fixed income securities. Added replay server test environment notation.	2012.03.21
2.2	Addition of BB Quoted flag to the Security Flag (3.3.6). Amendment of snapshot interval timing to every 10 minutes (from 3 minutes)	2012.02.21
2.1	Added specifications for snapshot request functionality. Added appendix section clarifying the tier to venue mapping.	2011.12.12
2.0	Added specifications for Ascii formatted Reference Data and Trade Data channels On Quote Flag field, renamed the QuoteType bit to QuoteSide.	2011.10.06
1.0.9	Made Security message compact by eliminating non-essential fields Added Number of marketmakers field to Inside and Price Level messages Added new value "SpinType=3" to Start of Spin and End of Spin messages	2011.08.12
1.0.8	Corrected offsets in Quote Add and Quote Update messages Specified that snapshots are sent every 3 minutes Marked the market open/close messages as non-optional Added restriction of max 2000 messages/request for Gap Fill	2011.07.28

	Requests	
1.0.7	Flipped Solicited to Unsolicited in Quote Flag Switched SecurityStatus from int to char in Security Message	2011.07.18
1.0.6	Removed trader message and also trader id field from Quote Add message Removed Rank fields from Quote Add and Quote Update messages Added Gap Fill Request and Response message definitions Included Ascii channel descriptions in Product/Channel mapping for completeness Added MessagingDisabledFlag in Security message	2011.07.08
1.0.5	Add market maker ID to quote message Full messages for add, delete and spin only all updates single sided	2011.06.20
1.0.4	Change name of Reference Message Types Remove bid and ask market maker ID from Quote Full Update Correct Action field values Correct Spin message order / contents	2011.06.17
1.0.3	Wrong field size in Price Level Reference Wrong data type in Trade Reference	2011.06.14
1.0.2	Sequence number to Spin and Market open/close	2011.06.03
1.0.1	Multicast recovery clarification Spin last sequence number addition	2011.05.25
1.0	Initial Version	2011.04.25